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Highlights

- Sole-author paper in *Finance Research Letters* (Scopus D1)
- Oral presentations at VPH 2026 and CMFF 2025
- 1st Place, EELISA international competition
- Two 1st places at BME research conferences

Skills

ML & Data

PyTorch, graph neural networks, image segmentation, Bayesian methods, SQL

UQ & Simulation

Polynomial chaos expansion, Gaussian processes, Dakota, VTK, PyVista

Programming

Python, MATLAB, C, C++, R, TypeScript, Mathematica, Solidity

Engineering

AutoCAD, Inventor, LabVIEW, MPLAB

Tools

Git, Linux, Slurm (HPC), Docker, Jupyter, L^AT_EX

Languages

Hungarian: native

English: C1 (IELTS 8.0)

German: B2

French: B1

Other

Driving licence (category B)

Interests

Ultra-endurance cycling
Drumming
PC hardware & overclocking
DIY – built a garden house

Damján MIKLÓS

Mechatronics Engineering BSc student

Machine learning and uncertainty quantification for cardiovascular simulations

Education

BSc in Mechatronics Engineering

Sep 2023 – Jan 2027 (expected)

Budapest University of Technology and Economics (BME)

GPA: 4.7/5 · Advanced Mathematics Track

Guest Student, ELTE Faculty of Science (Fourier Analysis university lectures during high school) 2023

High School Diploma, ELTE Trefort Ágoston High School 2017–2023

Research Projects

Uncertainty Quantification for Aneurysm Hemodynamics

Nov 2024 – present

Research Assistant, BME Department of Hydrodynamic Systems

- Developed an uncertainty quantification framework for aneurysm blood-flow simulations with polynomial chaos expansion (PCE) and Gaussian-process surrogate models (Dakota, Python, MATLAB)
- Derived a new semi-analytical method that constructs orthogonal PCE polynomials for truncated, correlated normal input distributions (unpublished)
- Built an efficient virtual patient sampling algorithm in MATLAB
- **Oral presentations** at VPH 2026 (Virtual Physiological Human), Milan, and CMFF 2025, Budapest
- **1st Place**, EELISA Scientific Student Competition 2026, Madrid (international)
- **1st Place**, GHK Special Award, BME Scientific Students' Conference (TDK) 2025 · Fluid Dynamics section

Generative Graph Neural Network for Carotid Artery Anatomies

2025 – present

BSc thesis, BME Department of Hydrodynamic Systems

- Developing a conditional generative graph neural network that creates anatomically accurate internal carotid artery (ICA) meshes with aneurysms from a single condition vector
- Built the cleaning and preprocessing pipeline for high-resolution ICA meshes in Python (VTK, PyVista)
- Developing a neural network for segmenting calcified ICAs
- **xxx Place**, BME TDK 2026 · Neural Networks section

Irrationality in Skewness Preferences

2024 – 2026

Independent research, BME Department of Finance

- Built a quasi-Monte Carlo simulation of hypothetical skewed return distributions, deriving a model-independent upper bound on the skewness premium from tail-risk measures
- Showed that the premia reported in six empirical studies exceed this bound at realistic horizons, pointing to behavioural drivers; [code](#) released open source
- **Published:** Miklós, D. (2026). *Finance Research Letters*, 109, 110514. doi:10.1016/j.frl.2026.110514 – sole author, Scopus D1 journal
- **1st Place**, Morgan Stanley Award, BME TDK 2024 · Finance section
- **3rd Place**, National Scientific Students' Conference (OTDK) 2025

Other Experience & Projects

Online Math & Physics Tutor Superprof

2023 – present

2000+ hours of university and high school-level tutoring in English and Hungarian; among the top-rated tutors nationwide

DynamicSpeed for YouTube

2026

Browser extension for Chrome and Firefox

Matches playback speed to a target words-per-minute estimated from captions